NanoCloud01: Data-Aware Continuation Training for a 99M-Parameter Language Model

Yirui Miao **Fengfei Yu** **Tianyue Zhang** **Zavier Zhao**
CSE 151B/251B, University of California San Diego

Abstract

We study autoregressive language modeling under a strict 100M-parameter limit. We compare three decoder-only designs that allocate capacity across depth, width, and grouped-query attention (GQA), then continue training the 99.03M-parameter Model 2. All models use RoPE, RMSNorm, SwiGLU, and tied embeddings. Model 2 is trained on a 50/20/15/15 mixture of FineWeb-Edu, Wikipedia, science, and books with Muon and AdamW. Its first epoch reaches 21.163 validation perplexity after 38k updates; low-learning-rate second-epoch continuation reaches **18.768** at step 69.7k, an 11.3% relative reduction. At this scale, data composition, optimizer choice, and checkpoint selection rival architecture.

1 Introduction

The NanoGPT competition, based on the compact training framework of Karpathy [3], asks for next-token prediction over the GPT-2 vocabulary while constraining the submitted model to at most 100M parameters. For a token sequence $x_{1:T}$, we minimize the following objective:

$$\mathcal{L}(\theta) = -\frac{1}{T}\sum_{t=1}^{T}\log p_\theta(x_t \mid x_{<t}), \qquad \mathrm{PPL} = \exp(\mathcal{L}). \tag{1}$$

The constraint makes capacity allocation unusually consequential: vocabulary embeddings already consume a large fraction of the budget, while depth, attention width, and feed-forward width compete for the remainder. Training must also generalize across the heterogeneous domains represented by the provided validation set rather than any single training source.

Our final approach combines a modern compact Transformer with deliberately balanced data and conservative continuation training. The architecture draws on rotary position embeddings [6], RMSNorm [7], SwiGLU [5], and grouped-query attention (GQA) [1]. Unlike larger-model work that emphasizes new modules, our strongest gain comes after the architecture is fixed: a second epoch resumes the first-epoch checkpoint with progressively smaller learning rates. Dense evaluation then captures a narrow validation minimum near convergence during late-stage training.

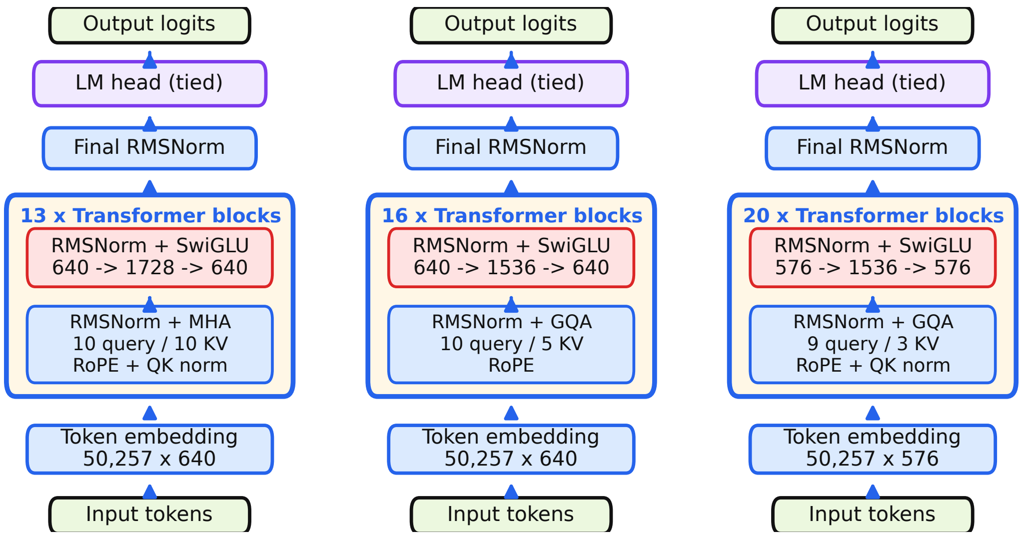
Contributions. First, we compare three sub-100M decoders that trade width and key/value capacity for depth. Second, we train Model 2 on a reproducible four-source mixture and show that balanced domains outperform single-source training. Third, we improve validation perplexity from 21.163 to 18.768 without changing the model or objective, demonstrating the value of low-rate continuation and checkpoint selection under a realistic training budget.

Preprint. Under review.

2 Method

2.1 Model Family

All three candidates are pre-normalized decoder-only Transformers with a 1024-token context, GPT-2 BPE, RoPE, RMSNorm, SwiGLU, bias-free projections, zero dropout, and tied input/output embeddings. They differ in depth, hidden width, feed-forward width, and key/value sharing; Figure 1 shows the resulting allocations and their depth–sharing trade-off directly.



(a) Model 1: 13L, 96.64M. (b) Model 2: 16L, 99.03M. (c) Model 3: 20L, 99.78M.

Figure 1: Model families under 100M parameters: MHA for Model 1, 1:2 GQA for Model 2, and 1:3 GQA for Model 3, enabling progressively greater depth under the shared parameter cap.

Model 1 uses 13 layers, hidden size 640, 10 query/key/value heads, a 1728-wide MLP, and QK normalization. Model 2 retains width 640 but uses 10 query and 5 key/value heads; the saved attention parameters support 16 layers with a 1536-wide MLP, totaling 99,032,320 unique parameters. Model 3 is the deepest design, with 20 layers, width 576, 9 query and 3 key/value heads, QK normalization, and a 1536-wide MLP. We continue Model 2 because its complete optimizer, loader, and random-number states were available at the end of the controlled 38k comparison.

The comparison tests three capacity-allocation strategies. Model 1 spends more parameters on independent key/value projections, Model 2 uses GQA to add three layers at the same hidden width, and Model 3 combines stronger GQA with a narrower residual stream to reach 20 layers. Models 1 and 3 also use QK normalization, whereas Model 2 does not. Because several dimensions change together, this is a model-level comparison rather than a component-wise causal ablation.

2.2 Data and Optimization

All sources are tokenized with GPT-2 BPE and stored as `uint16` binary arrays or memory-mapped shards. Every training example is a contiguous 1025-token window, split into 1024 inputs and their one-token-shifted targets. The loader follows an exact 20-example cycle: 10 FineWeb-Edu windows, 4 Wikipedia windows, 3 science windows, and 3 book windows. The cycle is reshuffled after each pass, while source cursors advance sequentially without overlap. FineWeb-Edu is the largest source [4]; the exact 50/20/15/15 mixture is shown in Table 1.

Training uses bf16 autocast, gradient clipping at 1.0, micro-batches of 16 sequences, and a global batch of 524,288 tokens per update. We assign two-dimensional hidden-layer matrices to Muon and embeddings plus remaining parameters to AdamW, following the small-model training practice popularized by modded-nanoGPT [2]. The first-epoch script warms up for 200 steps to peak learning rates 1.3×10^{-2} (Muon) and 5.2×10^{-4} (AdamW), then cosine decays to 10% of each peak by step 38k. It uses only the cross-entropy objective in Equation (1); no teacher or auxiliary loss is present.

The second-epoch script resumes the complete first-epoch state, including model, optimizer, loader, and random-number states. It keeps the architecture, data, batch, and objective unchanged, but lowers the effective Muon rate from approximately 3.0×10^{-4} at step 38k to 3.0×10^{-5} near step 69.7k; the AdamW rate remains 4% of the Muon rate. We branch from strong checkpoints at 62k, 63.6k, and 69.6k to shorten each cosine schedule and evaluate more frequently near convergence.

3 Experiments

3.1 Evaluation Protocol

We use the course evaluator on `val.bin`. It computes token-weighted cross entropy over 1024-token chunks and reports its exponential. Each evaluation covers 5,169,152 target tokens. The first epoch evaluates every 2k steps; the second increases frequency from 500 steps to 100 and finally 50 steps. The checkpoint at step 69.7k represents 36.54B sampled training tokens, of which 19.92B belong to the first epoch. All runs use PyTorch mixed precision on a single NVIDIA L40S 48GB GPU.

3.2 Ablations

Table 1 isolates data, architecture, and optimization. The mixture slightly beats FineWeb-Edu and decisively beats OpenWebText. Model 2 improves on Model 1 by 0.162 PPL, while Model 3 gains another 0.245. We continue Model 2 because its full training state is resumable. Muon + AdamW lowers PPL from 39.973 to 36.540, an 8.6% relative gain over AdamW alone in the controlled optimizer comparison. For Model 1, raising FineWeb from 50% to 53% and 56% also worsens PPL from 21.325 to 21.460 and 21.609. This supports balanced coverage rather than increasing FineWeb under the shared comparison budget.

Table 1: Ablation PPL (lower is better).

Comparison	Setting	PPL
Data source	FineWeb-Edu	39.67
	OpenWebText	65.21
	50/20/15/15 mix	**38.83**
Architecture	Model 1: 13L, MHA	21.325
	Model 2: 16L, GQA 1:2	21.163
	Model 3: 20L, GQA 1:3	**20.918**
Optimizer	AdamW	39.973
	Muon + AdamW	**36.540**

3.3 Main Result

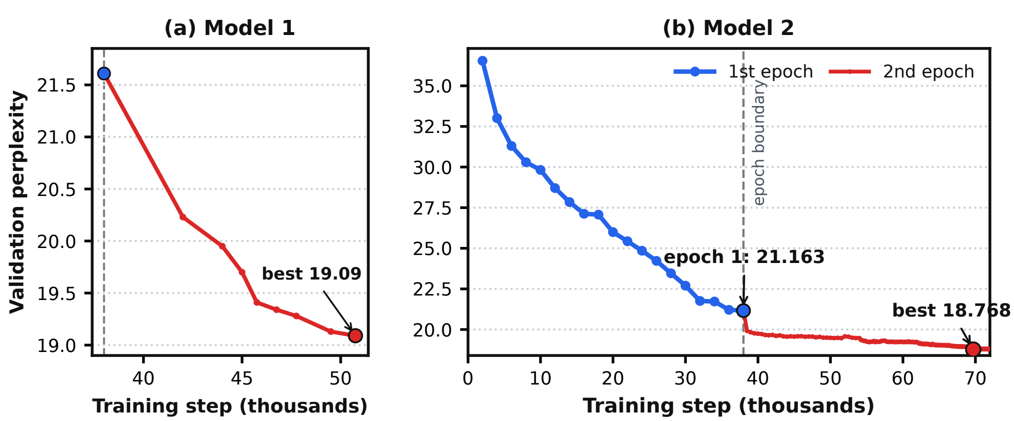


Figure 2: Low-rate continuation for Models 1 and 2. Red trajectories begin at the blue first-epoch endpoints, so each curve is a continuous checkpoint lineage. Model 1 values are team-reported; Model 2 values are verified from training logs and official evaluations.

Figure 2 shows consistent continuation gains across architectures. Model 1 improves from 21.609 to 19.09 by step 50.75k, an 11.7% relative reduction. Model 2's first epoch supplies the large representation-learning gain, reaching 21.163 at 38k; continuation reaches 19.666 at 42k, 19.210 at 62k, and 18.768 at 69.7k. Checkpoints through 71.8k then fluctuate around 18.79–18.81, making dense evaluation part of the method rather than only a reporting convenience.

The scales of the gains differ substantially. From step 2k to 38k, the first epoch reduces PPL by 42.1% relative; continuation then contributes another 11.3% from the 38k checkpoint. Model 3's 0.245-PPL advantage at 38k is much smaller than Model 2's 2.395-point continuation gain, although unequal training budgets prevent a direct ranking after 38k.

The curve also reveals that a single uninterrupted decay is suboptimal near convergence. Restarting from 62k and 63.6k with smaller schedule scales recovers steady progress, while the final 69.6k branch resolves a narrow minimum. These are continuation runs, not separate models: every branch inherits the same weights, optimizer moments, and data-loader position.

4 Discussion

What worked. The clearest result is that a fixed 99M architecture still has substantial headroom after its main schedule. Balanced multi-domain data avoids the severe failure of OpenWebText-only training, Muon accelerates early optimization, and low-rate continuation improves the converged checkpoint by 11.3% relative. The architecture comparison supports depth over redundant key/value capacity, but the schedule contributes the largest gain within Model 2.

Interpretation. The best 38k architecture is not automatically the best use of total compute. The model-level ablation favors Model 3 under a matched first-epoch budget, whereas Model 2 benefits more from a preserved training state and controlled continuation. Future comparisons should match both sampled tokens and checkpoint-selection effort, not only parameter count.

Limitations. The architecture comparison is not a complete factorial study: Models 1–3 have different depths and GQA ratios, and only Model 2 receives the full second-epoch continuation budget. Validation-driven branching also risks over-specializing to `val.bin`; the final hidden-test result is needed to establish distributional robustness. Finally, manual branch naming and evaluation failures at several checkpoints make experiment tracking more fragile than a single automated sweep.

Next steps. Future work should allocate equal token budgets to Models 1–3, automate best-checkpoint retention, and reserve a second held-out mixture for schedule selection. A small sweep over source weights around 50/20/15/15 would test whether the current optimum is robust. Under the same parameter limit, additional depth should be compared against narrower embeddings using matched throughput, rather than parameter count alone, in future controlled work.

Team responsibilities. Fengfei Yu developed and ran the two-epoch Model 2 training and evaluation pipeline and consolidated its results. **[Team input required: confirm Yirui Miao, Tianyue Zhang, and Zavier Zhao's responsibilities across Models 1/3, data ablations, slides, and report writing; also verify final leaderboard and repository details before submission.]**

LLM usage. An LLM organized local logs, edited prose, generated plotting code, and reviewed layout. Claims were checked against source code, evaluation logs, exported configurations, or the team presentation; the authors remain responsible for every result, interpretation, final wording, and submitted artifact carrying their names after independent human review and approval.

References

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A Reproducibility and Submission Checklist

Table 2: Evaluation points used for the ablations in Table 1.

Comparison	Training step
Data source	5,000
Architecture	38,000
Optimizer	2,000
Model 1 source-ratio sweep	38,000

Item	Location or value
Main training	`build-nanogpt/train_gpt2_v13.py`
Continuation	`build-nanogpt/train_gpt2_v16.py`
First-epoch checkpoint	`build-nanogpt/log/v13/train_ckpt_38000.pt`
Best exported model	`build-nanogpt/log/v16/.../submission_..._69700`
Best official validation	Step 69,700; PPL 18.7681466; loss 2.932161 nats
Required before submission	**[Add hidden-test PPL/rank, public code/model URLs, and confirmed team responsibilities.]**